

Supplementary Information for
When do single-cell response distributions improve drug retrieval?

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This Supplementary Information describes the methods, additional analyses and results supporting the main text. It contains Supplementary Methods, four Supplementary Notes and five Supplementary Tables. The supplementary analyses follow the same conceptual structure as the main text: they examine the information retained by population-level retrieval, the dependence of retrieval gains on the evaluation criterion, and the conditions under which state-specific single-cell responses can become consequential for candidate ranking.

10 Supplementary Methods

11 **Calculation of population retrieval scores.** Energy distance was the primary population score. RBF-
12 kernel maximum mean discrepancy used the median squared pairwise distance of the pooled query
13 and candidate sample as its bandwidth reference and evaluated scales of 0.25, 1 and 4 times that value.
14 Sliced Wasserstein-1 distance averaged quantile-matched projections over 128 fixed random directions.
15 Candidate populations were subsampled to at most 500 cells in the observed-population analyses and
16 300 cells in the partial-observation analyses. A stricter task-specific cap takes precedence where reported
17 in the main Methods. For a score with task seed s , the candidate and query populations were sampled
18 with seeds s and $s + 1$, respectively. Analyses without replicate-specific task seeds used $s = 0$; HIR-Bench
19 used $s \in \{0, \dots, 19\}$ and partial observation used $s \in \{0, \dots, 4\}$. A population containing fewer than two
20 cells after filtering was assigned distance 10^6 rather than removed from the candidate library. The same
21 value is used for an empty or undersized candidate component in the coverage score. Mean cosine and
22 mean L_2 first reduce each treated population to its mean and subtract the corresponding matched control
23 mean; query and candidate controls may differ. Energy distance, RBF-MMD, sliced Wasserstein and
24 coverage receive raw normalised-expression cell populations and do not subtract matched controls in the
25 primary population comparisons. Energy and RBF-MMD use the unbiased U-statistic by default; sliced
26 Wasserstein has no U/V choice, and coverage uses the same U default for its energy/MMD component
27 distances. The secondary PCA-latent population result uses the V-statistic form.

28 **Calculation of subpopulation coverage.** In construction-labelled experiments, the known construction
29 labels are used directly when they are carried by the task. In the controlled-mixture ranker, candidate la-
30 bels are unavailable, so candidate cells are assigned to the nearest target-component mean by Euclidean
31 distance. For partial observation, query cells are partitioned by k -means with $k = 2$, three initialisations
32 and the task seed $s \in \{0, \dots, 4\}$; candidate cells are assigned to the query component whose centroid has
33 the highest cosine similarity after row normalisation. A candidate with no cell assigned to a query com-
34 ponent received distance 10^6 for that component. At small β , the coverage distance was evaluated using
35 $D_\beta = \text{mean}(d) + \frac{\beta}{2} \text{var}(d) + O(\beta^2)$ to avoid numerical cancellation in the log-sum-exp form. Coverage-
36 mean and coverage-worst use the $\beta \rightarrow 0$ and $\beta \rightarrow \infty$ limits, respectively. The base distance was energy
37 distance unless otherwise specified; the complete score registry is given in Supplementary Table 1.

38 **Secondary retrieval summaries.** In the observed-population task, query-weighted Hit@1 was 0.884 for
39 energy distance and 0.421 for mean cosine, compared with the unweighted task-setting summaries re-
40 ported in the main text. PCA-space population and mean retrieval reached 0.778 and 0.518, respectively.
41 The both-mode candidate had no tied score at any rank across 1,260 queries and nine scorers, so opti-
42 mistic, pessimistic and fractional tie handling gave the same Hit@1. The CMap-style cosine and mean-
43 cosine implementations agreed exactly across 54,180 query-candidate scores (Spearman $\rho = 1.000$);
44 WTCS gave Hit@1 of 0.4635.

45 For MoA recovery, excluding the 21 leave-drug-out queries without a same-mechanism candidate left
46 the median population-minus-mean difference at 0.000 and changed the mean from -0.024 to -0.025 .
47 The decreased and increased shares changed from 40.8% and 35.7% to 42.3% and 37.0%, respectively.
48 This comparison uses worst-case coverage against mean cosine.

49 Supplementary Note 1. Query-level predictors of population benefit

50 The main text separates the existence of state-specific perturbation responses, their recoverability and
51 their effect on candidate ordering. Additional analyses evaluated whether simple observables could
52 identify queries for which population-level retrieval is more useful.

53 **Pre-specified query-selection rule.** The rule combined preference conflict with a structure-reliability
54 score. Preference conflict was $(1 - \bar{\tau})/2$, where $\bar{\tau}$ is the mean Kendall correlation between candidate
55 rankings computed separately for the two query states using energy distance. Structure reliability was
56 $0.15 \min(v/0.05, 1) + 0.10 \min(d, 1) + 0.40b + 0.35 \min(m, 1)$, where v is the fraction of query sum-of-
57 squares explained by a two-cluster partition, d is one minus the mean pairwise cosine similarity among

query cells, b is the fraction of five bootstrap resamples retaining the same top-ranked energy-distance candidate, and m is the Kendall distance between the mean-cosine and energy-distance rankings. All components lie in $[0, 1]$. Queries were selected when both scores were at least 0.4, assigned “mean sufficient” when structure reliability was at least 0.4 but preference conflict was lower, and otherwise assigned “mean or no call”. The response-matching and mechanism-recovery outcomes were not used to construct the rule.

Response matching and mechanism recovery across scores. Across the five population-level scores, median response-matching gains over mean cosine ranged from +0.028 to +0.067, whereas mean mechanism-recovery gains ranged from -0.033 to -0.011 across the 600 leave-drug-out queries. Thus, the divergence between the two evaluation criteria was not specific to the worst-case coverage score.

Alternative population scores in the observed-response task. Maximum mean discrepancy reproduced the energy-distance gain (MRR 0.975; +0.0101 over magnitude-aware mean L_2). Sliced Wasserstein reached 0.922, below directional mean cosine (-0.0230, 95% CI [-0.0301, -0.0166]) and magnitude-aware mean L_2 (-0.0431, [-0.0509, -0.0359]). The additional gain therefore depended on the population score.

Headroom and conditional associations. Directional mean cosine ranked the true drug first at every seed for 3,110 of 3,992 queries. Energy retrieval lost 0.0036 on average in these queries and gained 0.1501 in the remaining 882. The headroom-gain correlation ($\rho = 0.79$) was reproduced by the ceiling-preserving permutation null (0.786, interval [0.767, 0.805]), so it did not identify population-beneficial queries (Fig. 5d,e).

After conditioning on directional mean-cosine reciprocal rank, response detectability was associated with energy retrieval (+0.0091, 95% CI [+0.0060, +0.0121]; 3,618 queries in 44 cell-line clusters). Neither the cross-fitted interaction ratio (+0.0015, $p = 0.21$) nor its reproducibility (-0.0011, $p = 0.21$) showed a supported association. Replacing the cross-fitted measure with naive response cosine produced a negative association (-0.0096, 95% CI [-0.0167, -0.0024]), demonstrating sensitivity to how interaction is measured (Fig. 5f-h).

Control-reference sensitivity. We repeated the rankings after subtracting matched control means, keeping queries, candidate libraries, seeds and evaluation fixed. Across 2,250 SciPlex3 cross-line queries, energy Hit@1 changed from 0.9189 to 0.9158 and MRR from 0.9478 to 0.9462. All population-minus-directional-mean effects remained positive; Frangieh IFNGR1 contrasts were unchanged. A common control translation leaves the controlled-mixture, GDSC2 and same-condition RNA population distances unchanged.

Under partial observation, control subtraction also changed query clustering and candidate assignment. Across 765 queries, mean regret reduction increased from 0.1320 to 0.2021 for worst-case coverage and from 0.1155 to 0.2041 for mean coverage. The top candidate changed in 88.5% and 86.7% of queries, respectively. Thus, response-matching gains persisted, but individual rankings were reference-sensitive.

Recoverability of constructed response states. Supervised accuracy was 0.692 against 0.674 for the best unsupervised method in the class-pooled mixture, and 0.879 for individual drug-pair comparisons. Across nine alternative clustering configurations, median adjusted Rand indices ranged from 0.037 to 0.106. These comprised k -means with $k = 2$ in raw, 10-component PCA, 50-component PCA and response spaces; two-component Gaussian mixtures in raw and 50-component PCA spaces; and silhouette-selected $k \in \{3, 4, 5\}$, reported separately by the selected k .

In a sensitivity analysis over 20 seeds, increasing cell number from 25 to 400 per source changed the k -means adjusted Rand index by at most 0.02 at fixed separation. Increasing source separation from one to eight times its observed value raised the index from 0.07 to 1.00. Separation was rescaled using the full source populations before subsampling. More cells did not compensate for closely overlapping response states.

Measured response divergence. Queries were stratified by response divergence, defined as one minus the mean cosine between state-specific responses. Mean mechanism-recovery gains across increasing quartiles were -0.0497 , -0.0391 , -0.0158 and $+0.0142$. The divergence-gain association was positive ($\rho = 0.1548$, 95% CI $[0.0722, 0.2349]$), but the highest-quartile gain remained uncertain (95% cluster-bootstrap CI $[-0.0165, +0.0425]$; per-query standard deviation 0.169). Neither this stratum nor the full 600-query comparison established a consistent mechanism-recovery advantage (Fig. 3b,d).

Query-selection comparison. The rule assigns 127 queries to “mean or no call” and a further 11 to “mean sufficient”. Pooling the latter 11 with the unselected group leaves the comparison unchanged. Among individual-drug queries, the selected-minus-unselected median regret-gain difference is -0.0109 (95% drug-cluster bootstrap CI $[-0.0355, +0.0498]$; 584 versus 125 queries from 141 drugs). This comparison does not enrich for population-beneficial queries. The 11 mean-sufficient queries form a separate descriptive category.

Supplementary Note 2. HIR-Bench construction and biological transfer

HIR-Bench compares candidate preferences under different population objectives. The analytical threshold below describes the mean-welfare comparison between two specialist candidates.

Each synthetic query contains two subpopulations: a majority with mixture weight α and a minority with weight $1 - \alpha$. In a 50-dimensional latent space, each subpopulation has an orthogonal target response. Candidate response centres form a continuum from minority-specialist to balanced to majority-specialist, controlled by a conflict parameter in $[0, 1]$; remaining candidates are distant distractors. Candidate cells are Gaussian draws around their subpopulation-specific centres. The oracle utility of candidate d for subpopulation k is $u(d, k) = -\mathbb{E}\|X_{d,k} - t_k\|_2^2$, where t_k is that subpopulation’s target. Let d_M be the candidate with maximal utility in the majority and d_C the candidate with maximal utility in the minority. When $d_M \neq d_C$, define

$$A = u(d_M, 0) - u(d_C, 0), \quad B = u(d_C, 1) - u(d_M, 1).$$

The majority-weighted utility difference is $\alpha A - (1 - \alpha)B$, so its sign changes at

$$\alpha^* = \frac{B}{A + B}.$$

Thus α^* is the analytical boundary at which the majority-weighted preference switches between the two conflicting candidates; it is undefined and not evaluated when there is no preference conflict.

The full experiment crossed $\alpha \in \{0.50, 0.70, 0.85, 0.95\}$, seven equally spaced conflict levels from 0 to 1, 15 or 60 cells per subpopulation, noise standard deviation 0.3 or 0.8, candidate-library sizes 20 or 100, and 20 seeds ($s = 0, \dots, 19$), giving 13,440 synthetic instances across three candidate-information conditions. In the observed condition the original candidate populations are available. In the predicted-mean condition, each candidate population is replaced by Gaussian draws centred on its empirical mean, removing its original within-population structure. In the predicted-structure condition, each candidate population is reconstructed in a 10-component PCA basis and perturbed with Gaussian noise at 0.3 times the original noise standard deviation. Query populations and candidate libraries are otherwise fixed within each synthetic instance.

Query and candidate variance, isotropy and response diversity were used to predict the synthetic preference regime. Cross-validation grouped by the 28 $(\alpha, \text{conflict})$ combinations gave a pooled out-of-fold area under the receiver operating characteristic curve of 0.788; the majority class represented 64.3% of instances.

An exploratory transfer analysis included 239 tasks: 90 SciPlex3 within-context, 90 cross-context, 12 CD34+, 23 Frangieh and 24 predicted-mean tasks. A threshold fitted on 4,480 synthetic observed-population instances was transferred by matching percentiles of preference conflict. Accuracy was 39.7%, versus 95.8% for a constant classifier under the imbalanced labels (229 versus 10). These labels used the mean-based minority-state proxy that disagreed with the cell-level reference (Fig. 3c), so this analysis does not test transfer of a biological decision boundary.

150 HIR-Bench establishes constructively that within-population structure can become consequential for
151 candidate choice. The corresponding regime in biological data requires direct biological measurement.

152 **Supplementary Note 3. Robustness of the natural-heterogeneity and large-scale analyses**

153 The main text uses patient-derived glioblastoma and Tahoe-100M to ask whether the information re-
154 quirements identified in the controlled experiments are also present when cellular heterogeneity arises
155 within biological samples. These datasets measure differential response, recoverability and the extent
156 to which state-specific response ordering leaves headroom for population structure to alter candidate
157 preference.

158 **Response alignment between compartments.** For 17 patient–drug pairs in which both compartments
159 carried enough treated and control cells, the cosine between the malignant and myeloid perturbation-
160 response directions has a median of 0.566, against a range of 0.02 to 0.39 for the constructed mixtures
161 used elsewhere in this study. Read directly, that contrast says the two compartments of a real tumour
162 respond far more alike than two experimentally combined cell lines do.

163 The cosine is descriptive and depends on effect size, signal-to-noise and cell number. The two
164 natural-tissue results in the main text, the recoverability gap and conservation of candidate ordering
165 across disjoint compartments, are measured independently of this cosine.

166 **Robustness of glioblastoma compartment assignment.** The ZhaoSims2021 dataset does not provide
167 resolved cell-type labels for the analysed cells, so malignant, tumour-associated myeloid and oligoden-
168 drocyte compartments were assigned from marker scores. Cells with fewer than 200 detected genes
169 were removed, counts were library-size normalised to 10^4 and $\log(1 + x)$ transformed, and gene symbols
170 were upper-cased. Compartment scores were calculated on the full matrix before the 2,000-gene selec-
171 tion, so the labels do not depend on a gene set chosen for downstream variance. Three marker sets were
172 used: malignant *SOX2*, *EGFR*, *PDGFRA*, *OLIG1*, *OLIG2*, *GFAP* and *AQP4*; myeloid *PTPRC*, *CD14*, *AIF1*,
173 *CSF1R*, *ITGAM*, *C1QA* and *C1QB*; oligodendrocyte *MBP*, *PLP1*, *MOG* and *MAG*. A marker absent from
174 the matrix is skipped. Each gene was z-scored across all cells of the cohort and the compartment score
175 is the mean of its set’s z-scores, so the scores are cohort-dependent by construction. A cell was called for
176 the highest-scoring compartment only if the mean log-normalised expression of that compartment’s key
177 markers (malignant *SOX2*, *OLIG2*, *EGFR*; myeloid *CD14*, *CSF1R*, *C1QB*, *AIF1*; oligodendrocyte *MBP*,
178 *PLP1*) reached 0.25 and its z-score margin over the second-highest compartment reached 0.25. Cells fail-
179 ing either criterion were left unassigned, leaving 39.9% of cells unassigned. Every called compartment
180 was required to express its own key markers most strongly.

181 The results were stable when both thresholds were varied jointly over

$$\{0.10, 0.25, 0.40, 0.50\}.$$

182 The unsupervised arm is the best of four partitions of the same cells: k -means with $k = 2$, a two-
183 component Gaussian mixture, Leiden¹ and HDBSCAN². Leiden is the maximum on 152 of the 180
184 natural splits, so the reported best-unsupervised value is close to Leiden’s alone.

185 Across this range, the excluded fraction varied from 29% to 58%, while the supervised drug-response
186 ceiling remained between 0.915 and 0.923, the best unsupervised recovery between 0.77 and 0.80, and the
187 paired recoverability gap between approximately 0.11 and 0.13. The median cosine between malignant-
188 and myeloid-specific perturbation-response directions remained between 0.54 and 0.59 across thresh-
189 olds.

190 **Drug-response recoverability.** Within each patient and compartment, two treated populations were
191 balanced by sampling up to 150 cells per drug, requiring at least 50 per drug, across five seeds. Both
192 arms used the same up-to-30-component PCA fitted to the pooled, unlabelled mixture. Supervised ac-
193 curacy was estimated by five-fold stratified cross-validation of logistic regression, random-forest and
194 nearest-neighbour probes; scaling was fitted within each training fold. Unsupervised methods were

195 *k*-means, Gaussian mixtures, Leiden and HDBSCAN, evaluated by best-permutation accuracy. The pri-
196 mary comparison uses the best tested probe and clusterer; fixed-method comparisons assess sensitivity
197 to this selection. Supervised accuracy provides an empirical reference for recoverability.

198 **Patient-level uncertainty.** The natural-tissue recoverability analysis contains 36 within-patient drug-
199 pair splits across four patients and two compartments, with 30 of the 36 splits contributed by a single
200 patient. Drug-pair splits were therefore not treated as independent replicates. Patient-level cluster boot-
201 strap resampling was used to quantify uncertainty, and the analysis was repeated after removing the
202 dominant patient.

203 The reported recoverability gap is computed within each split before aggregation,

$$G_j = A_{\text{sup}}^{(j)} - A_{\text{unsup}}^{(j)},$$

204 not as the difference between the two marginal medians. The primary paired median is +0.117; sub-
205 tracting the marginal medians would instead give +0.147. Across 5,000 patient-cluster bootstrap re-
206 samples, the natural-minus-constructed difference in recoverability gap was +0.109 with an interval of
207 [+0.052, +0.129]. Four patient clusters contribute to this calculation, so the interval describes uncertainty
208 within this cohort.

209 Two aggregation controls give the same gap. Fixing every probe-clusterer pair, so that the same
210 pair is scored in both arms, gives +0.122 in tissue against -0.002 in the constructed arm. Removing the
211 patient that contributes 30 of the 36 splits gives +0.123, with a [+0.098, +0.138] interval.

212 **Large-scale measurement in Tahoe-100M.** Tahoe-100M plate 3 provides a substantially larger setting
213 in which heterogeneity occurs within cell lines rather than through constructed mixtures. Two cellular-
214 state definitions are used below, with partition rules specified independently of retrieval outcomes. The
215 first is the released cell-cycle phase call, G1 against G2M, with S-phase cells dropped as intermediate
216 and each state’s response referred to its own phase-matched control. The second is *k*-means with $k = 2$
217 fitted on each cell line’s DMSO_TF control cells in the 2,304 highly variable genes in log1p space, with
218 ten initialisations and seed 0; treated cells are assigned to the nearest control centroid, the larger control
219 cluster is taken as the first state, and a cell line enters only when it holds at least 100 control cells. Across
220 3,630 evaluable drug-cell-line conditions, the median cosine between the two state-specific response
221 directions was 0.739 (interquartile range 0.600–0.836). Fewer than 0.11% of conditions were as divergent
222 as the constructed mixtures used in the controlled experiments.

223 This estimate is not driven materially by the minimum-cell filter. Conditions were excluded when
224 one of the required state-specific treated or control arms contained fewer than 50 cells, which can prefer-
225 entially remove strong cell-cycle-arresting perturbations. Across drugs, coverage was only weakly asso-
226 ciated with induced-response cosine (Spearman $\rho = 0.077$), and restricting the analysis to well-covered
227 compounds changed the median from 0.739 to 0.740.

228 Recoverability remained measurable at scale. Across 960 sampled drug pairs and 48 cell lines, the
229 supervised ceiling was 0.853 and the best unsupervised procedure reached 0.611, giving a median paired
230 gap of 0.157. The gap was positive in all 48 evaluable cell lines. Individual unsupervised methods gave
231 similar results: *k*-means 0.592, Gaussian mixtures 0.582 and Leiden¹ 0.563.

232 **State-specific ordering and decision-relevance headroom.** We finally compared relative drug-
233 response ordering between non-overlapping cellular states. Under the cell-cycle partition, the median
234 disjoint Spearman correlation between the two drug-drug similarity structures was 0.841 across 44
235 evaluable contexts. Under the independently control-derived two-state partition, the corresponding
236 median was 0.778 across 45 contexts.

237 These high correlations indicate that relative drug ordering is often substantially preserved between
238 cellular states even when the states respond differently. The pattern is not universal: under the control-
239 derived partition, 2 of 45 contexts had a correlation below 0.5, with a minimum of 0.236. Such contexts
240 leave greater headroom for state-specific response information to alter candidate ordering. Headroom

of that kind is what a population-level method would need in order to change the selected candidate, and it is measured here on the ordering alone.

The two state definitions agree only moderately in their ranking of contexts (Spearman $\rho = 0.466$), reinforcing that apparent decision-relevance headroom depends on how cellular states are defined. Both partitions were specified independently of downstream retrieval performance.

Tahoe-100M therefore supports the same separation of concepts observed elsewhere in the study: state-specific responses can exist, and response identity can remain recoverable, while much of the relative candidate ordering is still preserved across cellular states. These are intrinsic-state diagnostics: they constrain the conditions under which population-level retrieval could add value. The advantage itself is measured on the same plate by the source-to-target intervention benchmark of the main text (Figs. 5 and 6).

Supplementary Note 4. Perturbation-predictor diagnostics

The predictor analyses ask whether a predicted response population contains state-specific perturbation effects that a downstream population-aware retriever could exploit.

Additive response models. For a predictor that applies the same perturbation effect δ_d to every cell in a context,

$$\hat{x}_{i,d} = x_i + \delta_d,$$

the response of any two cellular states is identical after subtraction of their corresponding controls,

$$r_{d,0} = r_{d,1} = \delta_d.$$

For a non-zero perturbation effect, consequently,

$$\cos(r_{d,0}, r_{d,1}) = 1.$$

This result is independent of the heterogeneity of the starting control population. An additive predictor can therefore generate a visually heterogeneous predicted population while containing no state-specific variation in the perturbation effect itself.

The average-effect and linear-latent predictors used in the main analysis satisfy this additive limit by construction when the perturbation response is referenced to the model-consistent baseline. They establish this limiting case.

Prediction-based retrieval in SciPlex3. This experiment was separate from the Tahoe-100M prediction task. Responses to each of ten divergent drugs were mixed across three cell-line pairs at three mixing fractions and eight seeds, giving 720 queries. Candidate responses were predicted from the corresponding mixed control populations and ranked by mean cosine, mean-delta R^2 , energy distance or coverage. Evaluation used nDCG@10 with drug identity as binary relevance. Average-effect, nearest-neighbour and linear-latent predictors used all SciPlex3 contexts and therefore provided in-sample references; entropic optimal transport excluded the query context. Sampling, library construction, model parameters and seeds are specified in Supplementary Table 4.

Separating perturbation effects from reconstruction error. Subtracting a model’s own zero-effect reconstruction isolates the induced response from reconstruction error. For an additive predictor $\hat{x}_{i,d} = g(x_i) + \delta_d$, this gives identical non-zero response vectors across states and cosine 1. Subtracting real controls instead also includes state-dependent reconstruction error.

Across 30 cell-line–drug combinations, induced-response cosines against the model baseline were 1.000, 1.000, 0.972 and 0.900 for average-effect, linear-latent, scGen and optimal-transport predictors, respectively. Against real controls they were 1.000, 0.190, 0.404 and 0.380; observed treated cells gave 0.205. The low real-control cosine of the additive linear-latent model therefore reflected reconstruction error, not a recovered state-specific perturbation effect.

282 **scGen.** scGen³ was fitted using scvi-tools⁴. A drug-specific latent shift was added to each encoded
283 control cell before decoding (Supplementary Table 4). Its induced-response cosine of 0.972 against its
284 own reconstruction was close to the additive limit, indicating less state dependence than observed in
285 treated cells.

286 **Entropic optimal-transport mapping.** We additionally evaluated an entropic optimal-transport map as
287 a predictor with cell-dependent displacements. It is a different map from the Gaussian (Bures) predictor
288 of the Tahoe-100M arm (Methods). Under the within-context differential-response analysis, the optimal-
289 transport predictions reached an induced-response cosine of approximately 0.900 ± 0.107 against their
290 own reconstruction baseline, moving away from the additive limit but remaining much more aligned
291 than the observed responses.

292 The entropic map was fitted using Sinkhorn optimisation⁵ in POT⁶. It used a 30-component PCA
293 representation, with the query context excluded from fitting; full parameters are given in Supplementary
294 Table 4.

295 In the cross-context predict-then-rank setting, the same entropic map gave Hit@1 values of 0.053
296 under mean cosine and 0.007 and 0.008 under the two distributional rules, compared with 0.879 to 0.899
297 for the additive average-effect candidate source. The predictions therefore did not preserve retrieval
298 performance in that transfer setting.

299 Taken together, these analyses identify a specific next experiment. A decisive test requires a pre-
300 dictor that both generalises accurately to the target cellular context and reproduces the state-specific
301 perturbation responses present in real cells. No predictor evaluated here satisfied both requirements
302 simultaneously.

Supplementary Table 1. Retrieval scores, task constructions and evaluation criteria.

(a) Retrieval scores.

Score	Input representation	Ranking rule	Role
Mean cosine	Control-referenced mean perturbation signatures	Descending cosine similarity between query and candidate mean-response vectors	Direction-only mean reference
Magnitude-aware mean L_2	Control-referenced mean perturbation signatures	Ascending Euclidean distance between query and candidate mean-response vectors	Mean reference retaining both response direction and magnitude
CMap-style cosine	Same control-referenced mean differential-expression vectors as mean cosine	Same cosine operation as the implemented mean-cosine baseline	Signature-retrieval reference; numerically identical to mean cosine under the specified scoring definition
WTCS	Mean-collapsed differential-expression response	Rank-based weighted connectivity / enrichment score	Canonical rank-based signature baseline; related to but not algebraically identical to mean cosine
Energy distance	Raw normalised-expression single-cell response populations	Candidates ranked by increasing U-statistic energy distance to the query population	Primary population-level retrieval score
RBF-MMD	Raw normalised-expression response populations	Candidates ranked by increasing U-statistic maximum mean discrepancy under an RBF kernel	Alternative distribution-sensitive score
Sliced Wasserstein	Empirical response populations	Candidates ranked by increasing sliced Wasserstein distance using 128 random projections	Alternative distribution-sensitive score
Subpopulation coverage	Response populations decomposed into response components	Component-level distances aggregated with temperature β ; increasing β gives progressively greater weight to the least well-matched response component	Population-sensitive score used for the partial-observation response-matching analyses

303 The four score families become five population-level configurations because subpopulation-coverage contributes two:
 304 coverage-mean is the $\beta \rightarrow 0$ limit of the aggregate and coverage-worst the $\beta \rightarrow \infty$ limit, both with energy distance as the
 305 per-component base metric and the same U-statistic default for energy/MMD components. The CMap-style cosine is numeri-
 306 cally identical to mean cosine on control-referenced mean vectors. Its sign-flipped variant is scored in the partial-observation
 307 experiments as an anti-correlation floor. Sliced Wasserstein has no U/V estimator choice; the PCA-latent population score uses
 308 the V form.

Supplementary Table 1 continued. (b) Retrieval task constructions.

Task	Query and candidate construction	Candidate evaluation	Statistical unit
Observed-population retrieval over heterogeneous queries	A heterogeneous query population combines two response modes at weight α ; the library holds a held-out re-mixture at the same weight, the two single-mode populations, and 40 distractors, giving a 43-candidate library	Hit@1; only the both-mode candidate is relevant, so a single-mode population counts as a miss even when it carries one of the query's two response modes	Retrieval query; headline values macro-averaged over seven task-setting combinations
Controlled heterogeneous queries	Two response modes are combined at mixing weight α ; controlled SciPlex3 analyses use HDAC- and JAK-associated response sources	Source identification and response-matching criteria	Query / task setting
Partial-observation retrieval	Only the response information available under the controlled partial-observation protocol is exposed to the retriever; every scorer, including direction-only mean cosine and magnitude-aware mean L_2 , ranks the identical candidate library once	The resulting fixed ranking is evaluated under response matching and MoA recovery; the mean-based minority-state similarity score is validated separately against a cell-level measure	765 total queries; formal inference clusters 720 individual-drug queries by 143 query drugs; 45 pooled partial-library queries are descriptive only
Leave-drug-out MoA analysis set	Same partial-observation rankings; all 600 leave-drug-out queries. Among these, 21 lack a same-mechanism candidate in the remaining library and receive $nDCG = 0$ for every method; they are retained as ties	MoA-nDCG	600 leave-drug-out queries from 130 query-drug clusters, including 21 zero-tie queries without a same-mechanism candidate; 120 leave-MoA-out and 45 partial-library queries are excluded because MoA recovery is undefined
Controlled-mixture validation of the state-similarity measure	K562 HDAC-JAK controlled mixtures at $\alpha = 0.7$, with 43 candidates under each of 20 fixed seeds	Mean-based minority-state similarity proxy versus a cell-level nearest-neighbour reference	860 candidate cases; top-ranked candidate compared within each seed
GDSC2-linked retrieval	Held-out SciPlex3 compound at $10 \mu M$ used as query; remaining matched compounds form a 30–34 evaluable-candidate library after valid-pair filtering	Spearman agreement between RNA-derived ranking and external GDSC2 functional-similarity ordering	103 query correlations pooled descriptively; cell-line-specific summaries retain the three higher-level units
Tahoe intervention identification (Figs. 5 and 6)	Vehicle cells and an observed treated population define each query; 92 candidates have either observed target-context responses or predictions fitted without that context	MRR and top-ranked intervention identity	3,992 queries in 44 cell-line clusters, each evaluated at five fixed seeds

Supplementary Table 1 continued. (c) Evaluation criteria applied to fixed rankings.

Criterion	Definition	Interpretation
Observed-population Hit@1 (Fig. 2)	Indicator that the both-mode candidate, the held-out re-mixture at the query's own weight, is ranked first	Recovery of the query's response composition, not of its perturbation identity alone

Supplementary Table 1(c) continued.

Criterion		Definition	Interpretation
Intervention-identification (Figs. 5 and 6)	MRR	Mean reciprocal rank of the query drug in the candidate library	Recovery of intervention identity from observed or predicted candidate responses
Response-matching regret		Energy-based welfare regret relative to the best available candidate, with utilities calculated separately for the response states of the controlled partial-observation task; reported relative to a named mean reference: direction-only mean cosine in Figs. 2c and 3a, and direction-only mean cosine or magnitude-aware mean L_2 separately in Fig. 2e	Response fidelity; intentionally close to the population information used by the retriever
MoA-nDCG		Normalised discounted cumulative gain rewarding candidates that share the query perturbation's annotated mechanism	Task-proximal biological criterion; 21 leave-drug-out queries without a relevant candidate were retained as zero-valued ties for all methods
Mean-based minority-state similarity proxy		Cosine between candidate-mean and minority-state-mean responses, rescaled to $[0, 1]$	Does not agree with the cell-level measure and is not used as a biological endpoint
GDSC2 functional similarity		Spearman correlation between cell-line-centred cross-cell-line AUC profiles of the query and candidate drugs	Externally measured functional similarity; not absolute potency
Protein direction-only evaluator		Cosine similarity between control-referenced mean protein-response vectors	Direction-only reference evaluator
Protein magnitude-aware mean evaluator		Negative Euclidean distance between control-referenced mean protein-response vectors	Primary mean-form evaluator matched to the Euclidean geometry of the population comparison
Protein population-energy evaluator		Negative energy distance between control-referenced per-cell protein-response populations	Primary population-form evaluator retaining cell-to-cell protein-response structure
Response-magnitude matching		$-\ \bar{d}_q\ - \ \bar{d}_c\ $	Query-dependent scalar control that uses no within-population information

Query states s are obtained by k -means with $k = 2$ on the query cells, the minority state being the smaller cluster. A candidate d with response population P_d receives the welfare $U(d) = \text{agg}_s(-\mathcal{E}(P_d, Q_s))$, where $\mathcal{E}$ is the energy distance and the aggregator is the minimum over states in the worst-case form and the mean in the mean form. The response-matching regret of a selected candidate $\hat{d}$ is $\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0$, taken over the observed library $\mathcal{L}$, so the reference is the best available candidate in the library. The mean-based proxy is $\frac{1}{2}(1 + \cos(\bar{P}_d, \bar{Q}_{\min}))$, the cosine between the candidate's mean response and the mean response of the query's minority state rescaled to $[0, 1]$. The same definitions and the cell-level validation measure are given in the main Methods.

Supplementary Table 2. Data resources, preprocessing and analytical roles.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
SciPlex3	276,325 cells; 2,000 genes	188 compounds shared across A549, K562 and MCF7	Log-transformed 2,000-gene representation; mean signatures are referenced to matched controls, whereas population scores use normalised treated-cell populations	Primary observed-population drug-retrieval setting; controlled within- and cross-context constructions; source of the GDSC2-linked transcriptomic rankings

Supplementary Table 2 continued.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
Frangieh Perturb-CITE-seq	218,023 RNA-profiled cells for retrieval and 218,331 barcode-matched cells for the RNA-protein evaluation; 2,000 genes; 20 retained surface proteins	239 CRISPR knock-outs across Control, IFN γ and Co-culture conditions	RNA and protein modalities joined by cell barcode; mean signatures are referenced to condition-matched controls, whereas population scores use normalised treated-cell populations	Independent genetic-perturbation retrieval setting and matched RNA-protein evaluator-shape experiment
ZhaoSims2021 glioblastoma	10 patients; 6 compounds	Acute-slice culture and surgical biopsy with co-existing malignant, myeloid and oligodendrocyte compartments	Counts normalised to 10 ⁴ per cell, log1p-transformed and reduced to 2,000 highly variable genes; state-specific responses referenced within patient and compartment	Natural-tissue analysis of differential response, recoverability and state-specific response ordering
CD34+ HSPC	33,984 cells; 2,000 genes	36-compound processed subset of GSE306429	Received as a processed expression matrix	12 exploratory proxy-based transfer tasks in Supplementary Note 2; not a primary retrieval endpoint
Tahoe-100M plate 3	4,158,278 cells; 2,304 genes	50 cell lines; 92 compounds at one concentration plus DMSO_TF vehicle controls	Released 2,304-gene log1p representation used without refitting; state-specific responses referenced to matched state controls	Two roles: large-scale measurement of differential response, recoverability and state-ordering agreement under intrinsic within-cell-line heterogeneity, and the substrate for the source-to-target intervention retrieval benchmark (Figs. 5 and 6)
GDSC2	286 compounds across 969 cell lines in the retained release	35 compounds matched to the SciPlex3 panel; A549, K562 and MCF7 excluded from external profiles	Fitted dose-response AUC; primary functional similarity uses cell-line-centred cross-line response profiles	External functional evaluation of fixed SciPlex3 RNA-derived rankings
HIR-Bench	13,440 synthetic instances	672 parameter configurations $\times$ 20 seeds in a two-subpopulation latent-welfare model	Synthetic observed, predicted-mean and predicted-structure information conditions	Controlled demonstration that population structure can alter candidate preference

316 The resources above address distinct questions. Dataset-specific cell-number thresholds, subsampling rules and statistical units
317 are given in the corresponding Methods sections.

Supplementary Table 3. Sensitivity of the GDSC2 functional-similarity evaluation to response-profile construction.

The primary external functional criterion compares drug–drug response profiles after removing the shared cell-line sensitivity effect. For cell line l and drug d ,

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in that cell line. Drug–drug functional similarity is then the Spearman correlation between centred cross-cell-line response profiles. The table compares this primary construction with the corresponding uncentred AUC-profile similarity.

RNA-derived ranking rule	Cell-line-centred functional similarity	Uncentred functional similarity
Energy distance	+0.265	+0.0902
Mean cosine, control referenced	+0.083	+0.138
Mean cosine, treated mean only	+0.241	+0.092
Response-magnitude matching	+0.232	+0.217
Candidate magnitude only	−0.330	+0.135
Potency matching diagnostic	+0.399	+0.290
Energy, magnitude-adjusted partial rank correlation	+0.105	n/a

Values are median per-query Spearman correlations over 103 leave-one-drug-out queries across the three SciPlex3 cell lines. The primary centred functional criterion is constructed from GDSC2 AUC profiles after subtracting each cell line’s mean response across all 286 compounds in the retained release. Drug pairs sharing fewer than 100 cell lines are excluded; 591 usable drug pairs remain, with a median of 909 shared cell lines per pair.

The change between centred and uncentred constructions shows that external evaluation is itself sensitive to how the functional measurement is summarised. Under the centred criterion, energy retrieval exceeds the control-referenced mean-signature baseline, but much of this association is also reproduced by response-magnitude matching. Under the uncentred criterion, the ordering between energy and the mean-signature baseline reverses. Neither construction is interpreted as a direct measure of therapeutic utility.

Supplementary Table 4. Perturbation-predictor implementations and their role in the differential-response analysis.

Predictor	Model form	Predicted-population construction	Use in this study	Interpretation / limitation
Average-effect	Per-drug mean perturbation delta	Predicted delta applied to the real control cells of the query context	Predict-then-rank candidate source and within-context differential-response reference	Additive by construction; state-specific induced-response cosine equals 1
Nearest-neighbour	Transfers the perturbation effect from the training context with the most similar untreated control state	Transferred effect applied to real query-context control cells	Predict-then-rank candidate source	Tests simple context transfer
Linear-latent	Additive latent arithmetic in an encoded representation followed by linear decoding	Observed control cells encoded, shifted in latent space and decoded individually	Predict-then-rank candidate source and additive latent reference	Can preserve baseline cellular heterogeneity while remaining additive in perturbation effect
scGen	Variational-autoencoder perturbation model with latent arithmetic	Evaluated on the same within-context state partition and state-specific response definition used for other predictors	Additional differential-response diagnostic reported in Supplementary Note 4	Mean induced-response cosine 0.972 over 30 cell-line and drug combinations

Supplementary Table 4 continued.

Predictor	Model form	Predicted-population construction	Use in this study	Interpretation / limitation
Entropic optimal-transport map	Non-parametric Sinkhorn mapping that permits cell-dependent displacements	Control cells mapped individually to predicted treated states	Additional non-additive differential-response diagnostic; also tested as a candidate-response source	Within-context induced-response cosine approximately 0.900 ± 0.107 against the map's own reconstruction baseline and 0.380 against the real control; cross-context candidate predictions were too inaccurate for a clean retrieval test

Supplementary Table 4 continued. Implementation of each predictor.

Predictor	Implementation
Average-effect	Per drug d , $\hat{\delta}_d = \text{mean}_c (\text{mean}(\text{treated}_{c,d}) - \text{mean}(\text{control}_c))$ over the SciPlex3 cell lines c , a line contributing only where it has at least two treated cells for d ; one signature per drug, independent of the query context. The predicted population is that single signature added to each of the query context's real control cells, so every cell is shifted identically. No reconstruction step, so the model's own zero-effect baseline is the control cells themselves. A drug absent from the fit is scored at -10^6 under every rule rather than imputed
Nearest-neighbour	$c^* = \arg \min_c (1 - \cos(\bar{x}_{\text{control}}^{\text{query}}, \bar{x}_{\text{control}}^c))$ over the training lines, then $\hat{\delta}_d = \text{mean}(\text{treated}_{c^*,d}) - \text{mean}(\text{control}_{c^*})$. Distances are compared strictly, so an exact tie is resolved by context iteration order; where d was not measured in c^* the signature falls back to the mean of d 's signatures over the remaining admissible lines. Population synthesis as for average-effect. The two query-composing cell lines remained eligible donor contexts, so this experiment is interpreted as an in-sample ceiling rather than as a transfer evaluation
Linear-latent	PCA with 30 components fitted, unsupervised and once, on up to 20,000 cells drawn from the full SciPlex3 matrix at seed 0. Each line's control latent mean is stored; each drug's latent delta is the mean over lines of (mean latent of treated – that line's control latent mean), again requiring at least two treated cells. Prediction encodes each real control cell of the query context, adds the drug's latent delta to that cell's own latent code, and decodes each cell by the PCA inverse transform. The decoder is linear, so the composite map is $\text{dec}(\text{enc}(x) + dz) = Px + Pdz$ and the predictor is additive in gene space and deterministic. Because it reconstructs, its own zero-effect baseline is the PCA reconstruction of the control cells rather than those cells
scGen	scGen fitted using scvi-tools; trained on the SciPlex3 expression matrix, perturbation as batch and cell line as label; 30 latent dimensions; batch size 512; at most 40 epochs with early stopping after 15 epochs without improvement; seed 0; per-perturbation latent delta added to each encoded control cell and decoded
Optimal transport	Entropic (Sinkhorn) coupling in a 30-component PCA latent; cost rescaled by its maximum; uniform marginals; regularisation 0.05; at most 500 iterations; fitted per non-query context centred on that context's own control mean, with the query context excluded; barycentric projection extended to the query's control cells by nearest-source matching and decoded by the PCA inverse transform

327 Predictors are used as candidate-response sources or information diagnostics rather than as a benchmark of perturbation-
328 prediction architectures. The average-effect, nearest-neighbour and linear-latent predict-then-rank diagnostics use the full
329 SciPlex3 matrix and are interpreted as in-sample candidate-generation references. The entropic optimal-transport diagnostic
330 instead excludes the query context during fitting and is assessed separately for cross-context transfer.

331 The predict-then-rank experiment used 720 cross-line queries, formed from the three cell-line pairs (K562–A549, A549–MCF7,
332 K562–MCF7) at mixing weights $\alpha \in \{0.5, 0.7, 0.9\}$ over 10 divergent drugs and 8 seeds; 200 query cells per query; a 31-drug
333 candidate library holding the true drug and 30 other shared drugs; 200 predicted cells synthesised per candidate; query con-
334 struction seeded at $1000 + s$ and distractor selection at $7000 + s$ for $s = 0, \dots, 7$. The within-context differential-response
335 measurements in Supplementary Note 4 are a separate run over 30 cell-line and drug combinations with 400 control cells per
336 context split into two states, a state requiring 60 cells and a treated arm 40, at seed 0. Nearest-neighbour appears only in the
337 predict-then-rank run and scGen only in the differential-response run; the entropic optimal-transport map appears in both.

Supplementary Table 5. Matched-geometry protein evaluator comparison.

RNA retrieval score	Protein evaluator		
	Direction-only cosine	Magnitude-aware mean L2	Population energy
All queries ($n = 659$)			
Direction-only cosine	+0.2527	+0.2967	+0.2299
Magnitude-aware mean L2	+0.1559	+0.4255	+0.3199
Population energy	+0.2235	+0.4150	+0.3930
Co-culture ($n = 219$)			
Direction-only cosine	+0.3132	+0.3087	+0.2760
Magnitude-aware mean L2	+0.2668	+0.4917	+0.3973
Population energy	+0.2666	+0.4771	+0.4412
Control ($n = 213$)			
Direction-only cosine	+0.2492	+0.2008	+0.1202
Magnitude-aware mean L2	+0.0657	+0.3177	+0.1544
Population energy	+0.2435	+0.3366	+0.3316
IFNγ ($n = 227$)			
Direction-only cosine	+0.2091	+0.3473	+0.2685
Magnitude-aware mean L2	+0.1521	+0.4426	+0.3394
Population energy	+0.1654	+0.4230	+0.3929

338 Entries are median per-query Spearman correlations across candidate knockouts. Rows are fixed RNA-derived retrieval scores
 339 and columns are evaluators constructed from the same held-out protein measurements. The primary matched-geometry com-
 340 parison is magnitude-aware mean-L2 versus population-energy retrieval under the protein mean-L2 and population-energy
 341 evaluators. The median within-query difference (RNA energy minus RNA mean-L2) is -0.0208 under the protein mean-L2
 342 evaluator and $+0.0538$ under the protein population-energy evaluator. It is -0.0105 and $+0.0466$ in Co-culture, -0.0363 and
 343 $+0.1342$ in Control, and -0.0318 and $+0.0384$ in IFN γ , respectively. Paired differences are medians of within-query differences,
 344 not differences between marginal medians.

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